

Development of a PPG Sensor Mounting Position Adjustment Device

Atsuhiko Fujii
Ritsumeikan University
Osaka, Japan

atsuhiko.fujii@iis.ise.ritsumei.ac.jp

Kazuya Murao
Ritsumeikan University
Osaka, Japan
murao@cs.ritsumei.ac.jp

Abstract

Pulse data is collected using a photoplethysmography (PPG) sensor, which is commonly found in wearable devices. The PPG sensor cannot obtain reliable pulse data if the sensor is loosely attached or positioned on a body part with fewer blood vessels. To address this issue, we propose a device that automatically adjusts the sensor's position when abnormal pulse data is detected. This device eliminates the need for manual intervention by the user and ensures the sensor maintains optimal placement, even in situations where adjustments are challenging, such as during sleep. We implemented a prototype device and collected three types of pulse data: the normal data obtained from a fixed PPG sensor as a baseline, the data obtained from a sensor on the device, and the data obtained from a sensor worn lightly wrapped with a band to reproduce a smart watch. Comparison of the obtained data showed that the pulse data obtained from the sensor on the device was closer to a normal pulse wave than the pulse data obtained from the sensor worn with a band. This result indicated that the proposed method of moving the PPG sensor to the optimal wearing position using the device is effective.

CCS Concepts

• Human-centered computing → Ubiquitous computing.

Keywords

pulse wave, heart rate, biometric information, PPG sensor, smart-watch, wearable robot, sensor position

ACM Reference Format:

Atsuhiko Fujii and Kazuya Murao. 2025. Development of a PPG Sensor Mounting Position Adjustment Device. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3714394.3756170>

1 Introduction

Wearable devices, such as smartwatches and smart rings, are gaining popularity, many of which can collect biometric information. One commonly used method for obtaining pulse wave and heart

rate data is photoplethysmography (PPG)[1]. PPG sensors operate based on the same principles introduced by Hertzman[6, 7], but they are designed in a smaller format using modern technology. These sensors work by emitting infrared, red, or green light onto the skin using LEDs. The oxidized hemoglobin in the blood flowing through the arteries absorbs these different wavelengths of light. PPG sensors take advantage of the fact that the amount of reflected light decreases as arterial blood flow increases with each heartbeat. By using a phototransistor, the sensors can detect changes in the amount of reflected light, allowing them to measure the pulse wave. The pulse wave data is represented as numerical values showing variations in reflected light, and the heart rate is calculated by identifying the peaks in this pulse wave data.

Due to their nature, PPG sensors struggle to collect accurate pulse wave data when they are worn loosely or placed on body parts with fewer blood vessels. This limitation is inherent to the PPG mechanism, making it difficult to address the core issue without properly reattaching the sensor. Various methods for denoising or completing missing pulse data have been proposed[2, 8, 11, 14]. However, these methods do not truly resolve the problem of missing data during collection, as they mainly involve estimating and compensating for the absent data rather than addressing the root cause. Adjusting the sensor mounting position when abnormal values are detected is essential to prevent the loss of raw data from PPG sensors. However, monitoring pulse wave data and regularly reattaching the device can be cumbersome for users, especially during sleep. We propose a device that automatically adjusts its mounting position when it detects abnormal pulse data. This automatic adjustment means that the user does not need to take any action. The proposed device ensures that the sensor remains in the correct position even during sleep, allowing for accurate data collection.

2 Related Work

We consider pulse data one of the most important types of biometric information, as it is frequently used to assess emotions and physical activity[5, 13, 17]. To use pulse wave data for state recognition, normal pulse data must be collected. Including abnormal pulse data may degrade the accuracy of state recognition.

There are numerous studies focused on optimizing the positions for mounting sensors. Ha et al.[4] achieved high-accuracy measurement and high-robustness operation in monitoring both body temperature and heart rate by optimizing the placement positions of biosensor arrays. Xu et al.[16] focused on optimizing the placement of Forcemycography sensors and proposed an optimization algorithm with greedy local search. Davoudi et al.[3] compared the accuracy performance between multiple and variable placements

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp Companion '25, Espoo, Finland

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3756170>

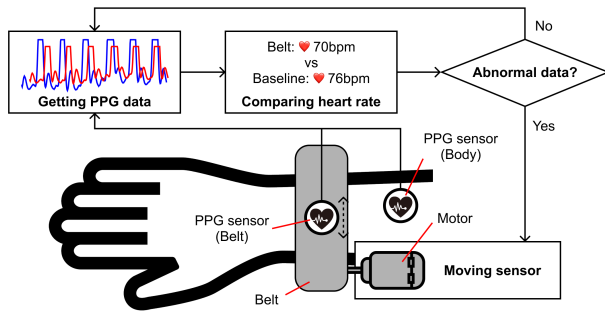


Figure 1: Process flow of proposed system.

of accelerometer devices in categorizing the type of physical activity and corresponding energy expenditure in older adults. Xia et al.[15] designed an optimization scheme with virtual sensor data for the Human activity recognition system. The sensor position must be carefully considered to obtain normal data.

There are several studies aimed at improving the quality level of the PPG signal by using multiple PPG sensors or photodiodes[9, 10, 12]. However, these methods cannot continue to obtain pulse data when all sensors are detached from the skin. We propose a device that automatically adjusts the wearing position of the PPG sensor so that it remains attached to the skin, even if it becomes detached, thereby enabling continuous data acquisition.

3 Proposed Device

This section describes the details of the proposed device.

3.1 Functionalities

The process flow of the proposed system is illustrated in Figure 1. The prototype device utilizes two PPG sensors: one is mounted on the belt, while the other is securely attached to the body to prevent misalignment and to gather baseline data. The proposed device is activated if it detects a disturbance in the pulse data obtained from the PPG sensors on the belt based on the baseline pulse data. This prototype device is implemented for evaluation experiments to validate the effectiveness of the fundamental mechanism of moving the sensor’s attachment position by the device. Ultimately, we think of using only the data obtained from the PPG sensor on the belt to determine if the pulse data is abnormal using a machine learning model.

The system collects data from two sensors simultaneously and compares the results. If the system identifies an abnormal waveform, a motor connected to the belt will activate to reposition the sensor. Conversely, if the system detects a normal waveform, the motor remains inactive.

3.2 Implementation

The sensor movement mechanism of the implemented prototype device is shown in Figure 2a. The mechanism is composed of a motor, a driving pulley, an idler pulley, a toothed belt, and a PPG sensor¹. Both the motor and the PPG sensor are controlled by an

¹<https://pulsesensor.com>

Arduino UNO R4 WiFi microcontroller. The motor driver used is a L298N module, and the external power supply provides DC power at 12V/1A. The motor’s rotation can be controlled by applying a HIGH (5V) or LOW (0V) voltage to the digital pins on the Arduino. If the polarity is reversed, the motor will rotate in the opposite direction. Additionally, setting both pins to HIGH activates the braking.

The PPG sensor is fixed to the toothed belt using a glue gun. The toothed belt has a buckle, so it can be worn around the wrist like a watch. The belt is set to pass over two pulleys so that the power of the pulleys is transmitted to the belt. When the motor is activated, the timing pulley connected to the motor rotates the belt, which in turn moves the PPG sensor with the belt. However, the sensor cannot pass over the pulley due to interference and cannot make a full revolution around the wrist. Therefore, by controlling the direction of rotation of the motor, the PPG sensor moves only on the back of the wrist.

3.3 Disturbance detection

The detection of disturbances in pulse data was implemented by comparing the heart rates of two PPG sensors. A disturbance is determined if the heart rate obtained from the PPG sensor on the belt differs by more than 5 bpm from the heart rate obtained from the baseline PPG sensor.

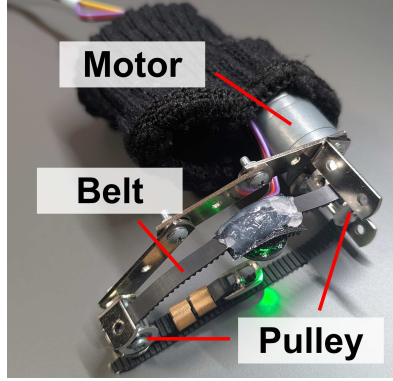
The heart rate calculation is performed every 5 seconds and is computed using Equation 1 based on the number of peaks P_{count} detected during that time.

$$HeartRate = \min(\max(P_{count} * 12, 30), 200) \quad (1)$$

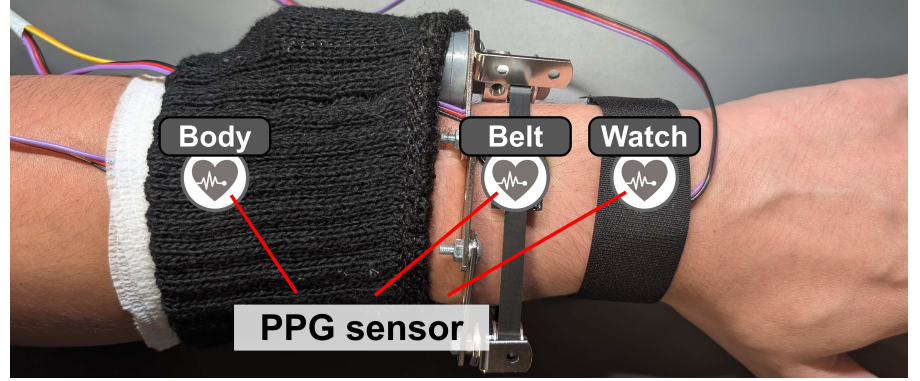
The PPG sensor provides an analog output of 0–1023 values. The heart rate is calculated only if the peak is detected at least once every 5 seconds and the latest sensor data is between 50–950. Otherwise, the heart rate is output as 0 beats per minute (bpm) because the sensor may be out of the body, resulting in anomalous values. The number of peaks in 5 seconds is multiplied by 12 to convert to 60 seconds, and the output is limited to a range of 30–200 bpm to exclude extreme values.

The most recent five values counted from the current time t obtained from the sensor are kept in $Buffer[i]$ ($i = t, \dots, t-4$). Peak detection is performed only when $Buffer[i]$ is full and the latest sensor value is between 50–950. The detection process follows this flow. First, it checks that $Buffer[t]$ is greater than the threshold T . Next, $Buffer[t]$, $Buffer[t-1]$, and $Buffer[t-4]$ are compared, and if $Buffer[t]$ is the largest, the peak is detected and P_{count} is counted up. However, a peak is not detected until 300 milliseconds or more have elapsed since the most recent peak detection. This condition was set because an interval of 300 milliseconds corresponds to 200 bpm when calculated as a heart rate.

The amplitude of the pulse data obtained from the PPG sensor varies depending on the wearing position and other conditions. Therefore, the threshold T for peak detection is set at 600 at the start, but is implemented to change dynamically on a moving average basis thereafter. The last 10 samples are used for the moving average, and the difference between the average and the latest value is calculated as the amplitude component. To emphasize this amplitude component, it is doubled and multiplied by a constant coefficient ($R = 0.4$). This value is added to the average value, and



(a) Overview of the sensor movement mechanism of the device.



(b) The device worn in the evaluation experiment.

Figure 2: Implemented prototype device.

the integer value is used as the threshold T . The constant coefficient R is a coefficient to determine the degree of application of the amplitude component to T , which was heuristically determined to be 0.4. If the last 10 pulse samples were $value[i]$ ($i = 0, \dots, 9$), the threshold T is calculated as in Equation 2.

$$T = \text{int}(\mu + |value[9] - \mu| \cdot 2R), \quad \mu = \text{mean}(value) \quad (2)$$

3.4 Motor control

The heart rates are compared in real time, and the motor waits until a disturbance in the pulse wave occurs. When a disturbance of the pulse wave is detected, the motor is activated to move the sensor position. Since it takes some time for the measured values to stabilize after moving the PPG sensor position, a 10-second calibration period is allowed before the next disturbance is detected. After 10 seconds, the system resumes disturbance detection and repeats activation and deactivation of the motor depending on the detection status.

The proposed device can move the PPG sensor toward the thumb or pinky finger, and put the sensor in any position on the wrist by controlling the direction of rotation of the motor. In this paper, the direction of rotation is heuristically set as a fixed value. The past directions of motor rotation are recorded, and the rotation is repeated in the order of [thumb direction, thumb direction, pinky finger direction, pinky finger direction]. However, the distance to be moved cannot be fixed with the current mechanism because the resistance applied to the belt varies depending on the wearing condition.

4 Evaluation

We conducted an evaluation experiment to verify the effectiveness of the fundamental mechanism of the device.

4.1 Data collection

The sensor and device were worn on the left wrist, and data was collected for 11 minutes during a seated, computer-based, mostly typing desk job. The raw data obtained from the PPG sensor were

recorded at a sampling rate of approximately 14 Hz. The PPG sensor was attached at the three points indicated as *Body*, *Belt*, and *Watch* in Figure 2b. *Body* is the sensor used to obtain the correct pulse wave, and was fixed with a bandage so that it would not move. *Belt* is the sensor implemented on the belt. *Watch* is a position that replicates a smart watch for the evaluation experiment, lightly wrapped with a band and connected to the microcontroller used on the prototype device. All sensors were worn near the left wrist to collect data at the same time and location as much as possible, and set to have the same sampling rate. The detection of pulse wave disturbances in the prototype device was achieved by comparing data from *Belt* and *Watch*.

4.2 Results and discussion

The distribution of the heart rates obtained from each sensor, rounded down to the nearest 10, is shown in Figure 3. Of the 11 minutes of recorded data, the first minute was excluded for calibration purposes, and only the remaining 10 minutes were used for analysis. Note that the heart rate calculation is performed and updated every 5 seconds, so that a constant sample is recorded for at least 5 seconds. *Body* is the correct pulse data obtained from the PPG sensor, and is the basis for evaluation. Since the data were collected while a subject was seated at a desk, the heart rate was stable, most of the samples were concentrated at 70–89 bpm. The heart rate distribution of *Belt* is also concentrated in the 70–89 bpm range, indicating that heart rates close to the correct pulse data were obtained. However, the number of samples was larger than that of *Body* in the range of 90 bpm. We think that this is due to the influence of arm movements. The fact that almost the same number of samples were recorded for *Belt* and *Watch* indicates that both may have been affected. If we check the range of 40, 130, and 140 where no samples are recorded on *Body*, we find that there are samples for *Watch*, but none for *Belt*. This indicates that *Belt* suppressed the effect of arm movements to some extent compared to *Watch*.

Table 1 shows the results of calculating several indices from all samples of heart rate for each sensor. The mean, minimum, maximum, median, and standard deviation of *Belt* were all closer to *Body* than to *Watch*. Exceedance Rate is the percentage of samples

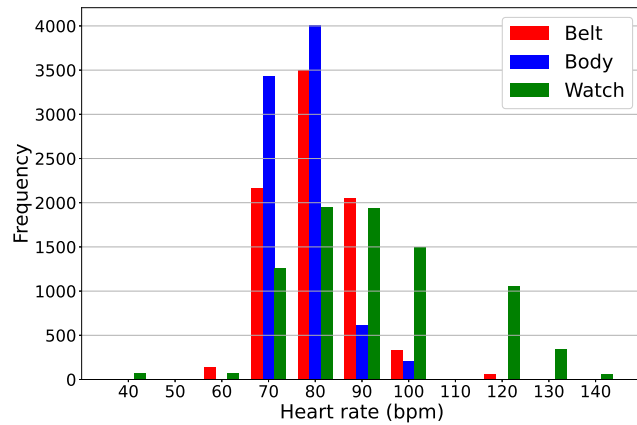


Figure 3: Distribution of heart rates obtained for each sensor.

Table 1: Summary of heart rates obtained for each sensor.

Sensor	Belt	Body	Watch
Mean [bpm]	84.7	80.5	95.9
Minimum [bpm]	60	60	48
Maximum [bpm]	120	108	144
Median [bpm]	84	84	96
Standard Deviation [bpm]	10.8	8.5	18.0
Exceedance Rate	45.1%	-	73.9%

with a difference of more than 5 bpm compared to *Body*, indicating that *Belt* was able to measure values closer to *Body*. These results indicate that more stable pulse data can be collected when the PPG sensor is worn with a device that adjusts the wearing position than when it is simply worn on the wrist. Therefore, the method of repositioning the sensor to achieve a normal waveform is effective for cases when the PPG data presents an abnormal waveform.

5 Future Work

In the future, we will develop a system that can detect disturbances based on raw PPG data rather than heart rate. When the PPG sensor is not properly attached, flat or random values without peaks may be measured. We will construct a machine learning model that can recognize abnormal pulse data based on features such as noise and integrate this capability into the motor control mechanism of the prototype device. The realization of this system provides a mechanism that allows the device to continue collecting normal pulse data on its own, without needing a baseline PPG sensor. We will also implement a mechanism that gradually adjusts the movement distance of the PPG sensor depending on the quality of the collected data. In addition, we will conduct user surveys on the comfort level of the prototype, such as its size and operating noise, with an increased number of subjects and in various dynamic tasks other than desk jobs.

6 Conclusion

We proposed a device that adjusts the sensor's attachment position when it detects abnormal pulse data. We built a prototype of this device and conducted evaluation experiments using data collected from the PPG sensor integrated into it. The results showed that when the PPG sensor was worn with the device and the wearing position was adjusted, more normal data was continuously obtained than when it was only lightly wrapped with a band. In the future, we plan to develop a machine learning model that can recognize whether the PPG data represents a normal or abnormal waveform based on raw PPG data. This model will be integrated into the motor control system of the prototype device.

References

- [1] John Allen. 2007. Photoplethysmography and its application in clinical physiological measurement. *Physiological Measurement* 28, 3 (feb 2007), R1–R39.
- [2] George R.E. Bradley and Panayiotis A. Kyriacou. 2024. Opening the envelope: Efficient envelope-based PPG denoising algorithm. *Biomedical Signal Processing and Control* 88 (2024), 105693.
- [3] Anis Davoudi, Mamoun T Mardini, David Nelson, Fahd Albinali, Sanjay Ranka, Parisa Rashidi, and Todd M Manini. 2021. The Effect of Sensor Placement and Number on Physical Activity Recognition and Energy Expenditure Estimation in Older Adults: Validation Study. *JMIR Mhealth Uhealth* 9, 5 (3 May 2021), e23681.
- [4] Seulki Ha, Sangheon Park, Hyeoncheol Lim, Seung Ho Baek, Do Kyoung Kim, and Sang-Hee Yoon. 2019. The placement position optimization of a biosensor array for wearable healthcare systems. *Journal of Mechanical Science and Technology* 33, 7 (01 Jul 2019), 3237–3244.
- [5] Yunjo Han, Kobijon E. Toshnazarov, Byung Hyung Kim, Youngtae Noh, and Uichin Lee. 2023. WatchPPG: An Open-Source Toolkit for PPG-based Stress Detection using Off-the-shelf Smartwatches. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*. 208–209.
- [6] Alrick B. Hertzman. 1938. Comparative Estimation of Blood Supply of Skin Areas from Photoelectrically Recorded Volume Pulse. *Proceedings of the Society for Experimental Biology and Medicine* 38, 4 (1938), 562–564.
- [7] Alrick B. Hertzman. 1938. THE BLOOD SUPPLY OF VARIOUS SKIN AREAS AS ESTIMATED BY THE PHOTOELECTRIC PLETHYSMOGRAPH. *American Journal of Physiology-Legacy Content* 124, 2 (1938), 328–340.
- [8] B. Keerthiveena and S. Esakkirajan. 2017. Denoising of PPG signal by wavelet packet transform. In *2017 International Conference on Intelligent Computing, Instrumentation and Control Technologies (ICICICT)*. 608–612.
- [9] Ahmad Khatoun, Adrian-Razvan Petre, Bart Peeters, Albert J. J. M. van Breemen, and Hylke B. Akkerman. 2023. Wearable Near-Infrared Organic Photodiode Arrays for Cardiovascular Health Monitoring and Biomarker Mapping: Signal-to-Noise Ratio Enhancement Using Machine Learning. In *2023 IEEE SENSORS*. 1–4.
- [10] Zi-Jun Liu, Ting Xiang, Ru-Shuang Zhou, Nan Ji, and Yuan-Ting Zhang. 2024. A multi-channel photoplethysmography array with contact-force regulation for tonoarteriographic imaging. *Connected Health And Telemedicine* 3, 1 (2024).
- [11] Bhanupriya Mishra and Neelam Sobha Nirala. 2020. A Survey on Denoising Techniques of PPG Signal. In *2020 IEEE International Conference for Innovation in Technology (INOCON)*. 1–8.
- [12] Jose Ignacio Rodriguez-Labra, Christopher Kosik, Dinesh Maddipatla, Binu B. Narakathu, and Massood Z. Atashbar. 2021. Development of a PPG Sensor Array as a Wearable Device for Monitoring Cardiovascular Metrics. *IEEE Sensors Journal* 21, 23 (2021), 26320–26327.
- [13] K Selvakumar, E Vinodh Kumar, M Sailesh, Mamtani Varun, Anbu Allan, Nanda Biswajit, Panga Namrata, and Sivaramkrishnan Upasana. 2022. Realtime PPG based respiration rate estimation for remote health monitoring applications. *Biomedical Signal Processing and Control* 77 (2022), 103746.
- [14] Hangsik Shin. 2024. Signal completion using generative adversarial networks for enhanced photoplethysmography measurement accuracy. *Computers in Biology and Medicine* 180 (2024), 108952.
- [15] Chengshuo Xia and Yuta Sugiura. 2021. Optimizing Sensor Position with Virtual Sensors in Human Activity Recognition System Design. *Sensors* 21, 20 (2021).
- [16] Xiaohao Xu, Zihao Du, Huaxin Zhang, Ruichao Zhang, Zihan Hong, Qin Huang, and Bin Han. 2022. Optimization of Forcemycography Sensor Placement for Arm Movement Recognition. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 9845–9850.
- [17] Zhibin Zhu, Xuanyi Wang, Yifei Xu, Wanlin Chen, Jing Zheng, Shulin Chen, and Hang Chen. 2025. An emotion recognition method based on frequency-domain features of PPG. *Frontiers in Physiology* Volume 16 - 2025 (2025).